Study design: preoperative muscle morphometry and non-home discharge after lumbar spine surgery

Cohort & segmentation

205 patients
51 non-home discharge (24.9%)

Preoperative lumbar MRI:
3D Slicer / TotalSegmentator v5.8.1, L3–L5

Per muscle:
L4-normalized volume + bilateral-mean T2 signal

Segmented muscle groups: Iliopsoas · Deep paraspinal (erector/multifidus) · Gluteus medius

Model specification

Outcome: non-home discharge (inpatient rehabilitation / skilled nursing facility vs home)

ADJUSTED FOR (clinical, pre-specified):
Age · Female sex · ASA class ·
No. of operated levels · Fusion vs decompression

MUSCLE PREDICTORS ADDED:
Iliopsoas, Deep paraspinal, Gluteus medius
(normalized volume + mean T2 signal)

M0 Clinical
5 predictors

M1 + Iliopsoas
7 predictors

M2 + Multi-muscle
11 predictors

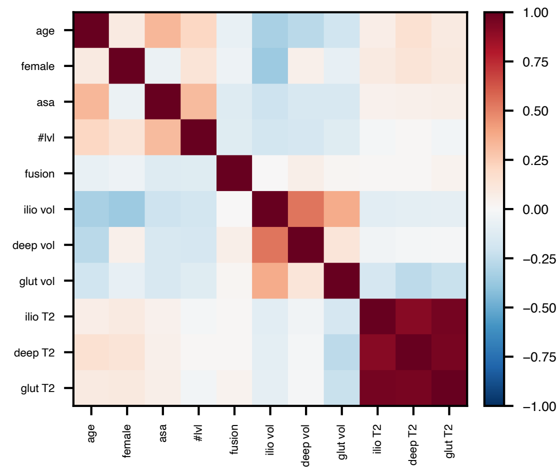
Multivariable logistic regression · predictors standardized to per-SD effects · ridge (L2) shrinkage · no data-driven selection · EPV = 51/11 = 4.6

Estimation & diagnostics

Leak-free repeated stratified 10-fold CV ×100
(impute + standardize INSIDE folds) + bootstrap optimism

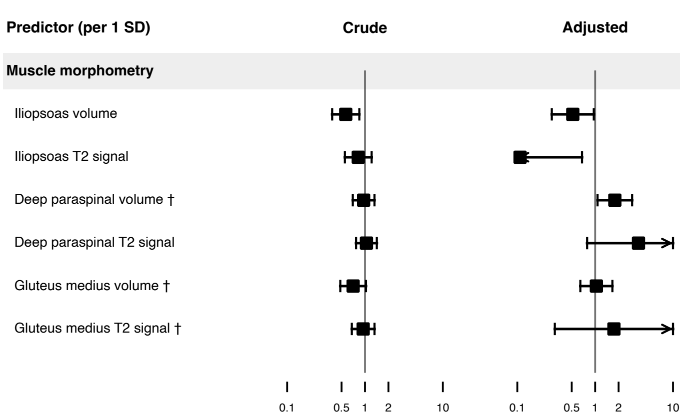
Univariable vs adjusted OR + VIF checked
for collinearity / suppression artifacts

Predictor correlation

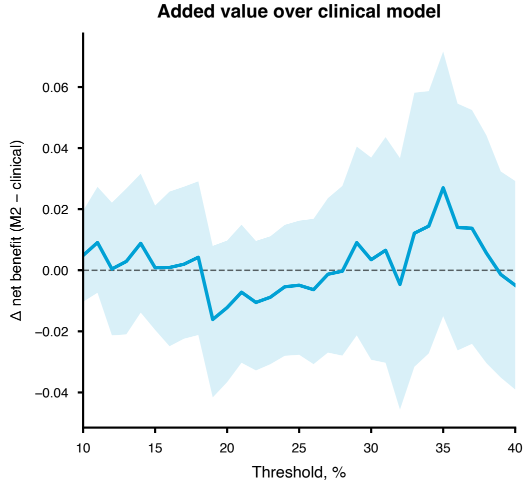


Findings

Muscle predictors — crude vs adjusted OR



Added value over clinical model



Primary finding: lower iliopsoas volume is independently associated with non-home discharge (OR per SD 0.52). Adding muscle morphometry does NOT improve prediction over clinical factors (Δ net benefit ≈ 0; P[>0]=0.59). † = collinearity/suppression artifact (not interpreted).